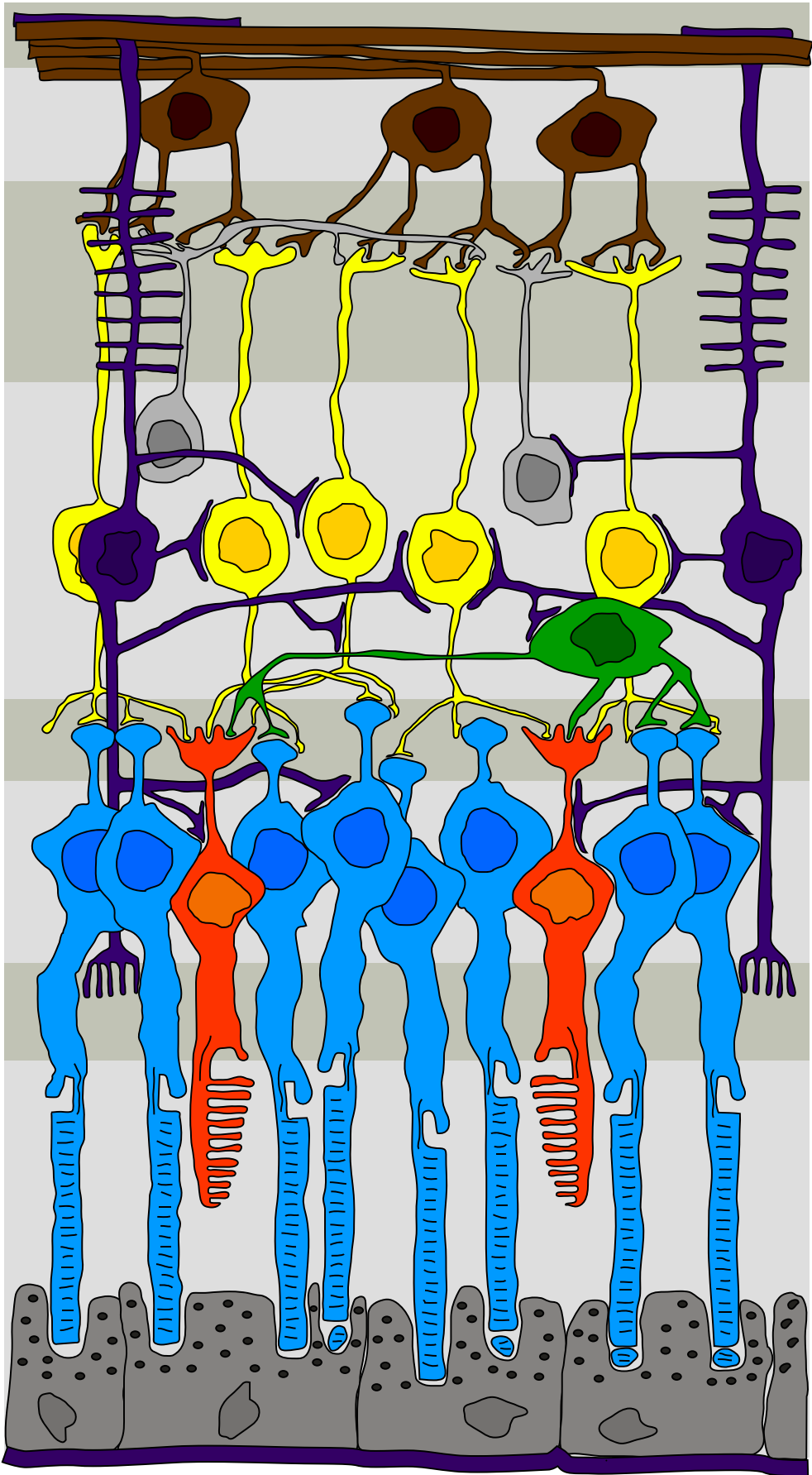


Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY

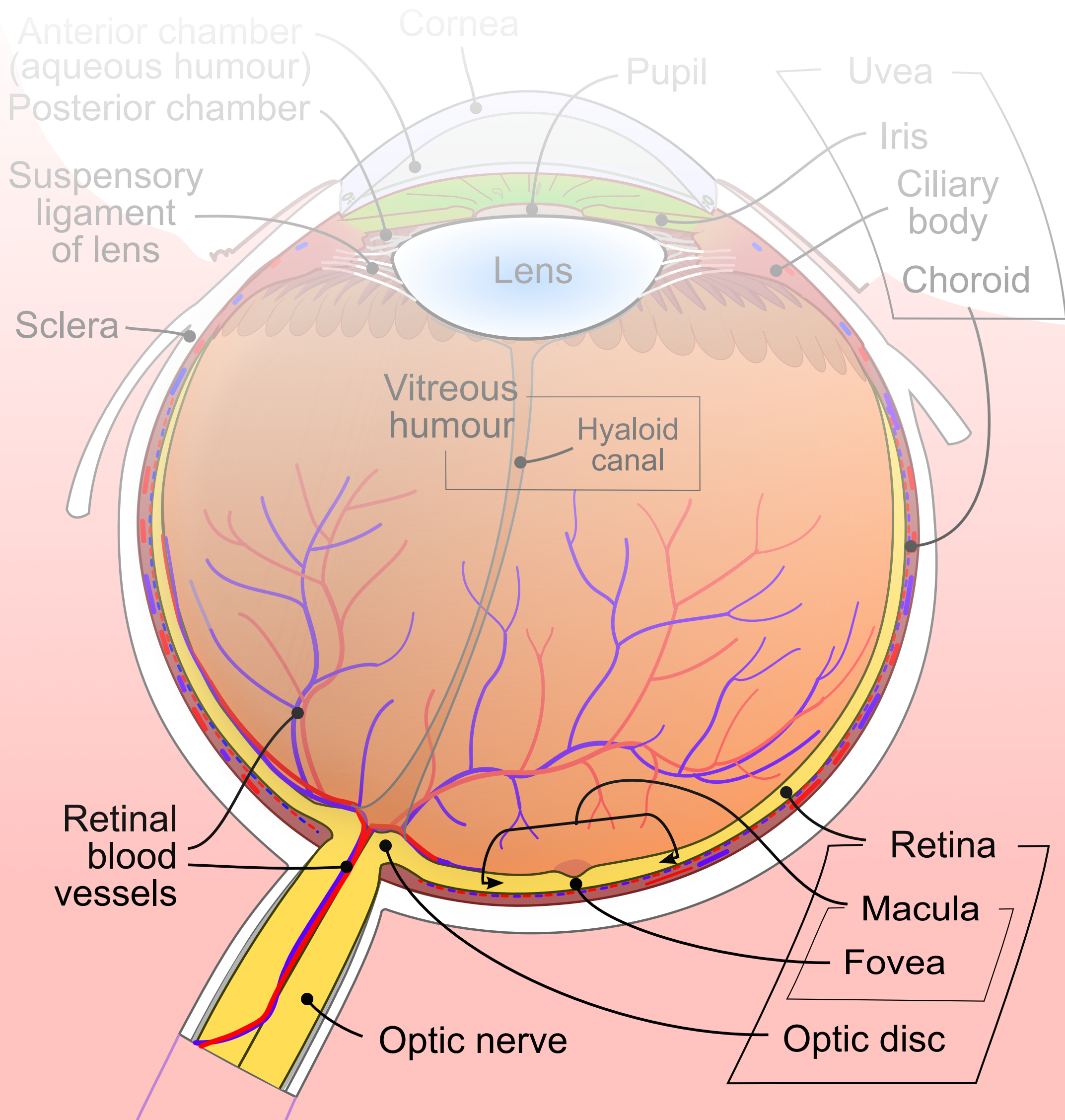


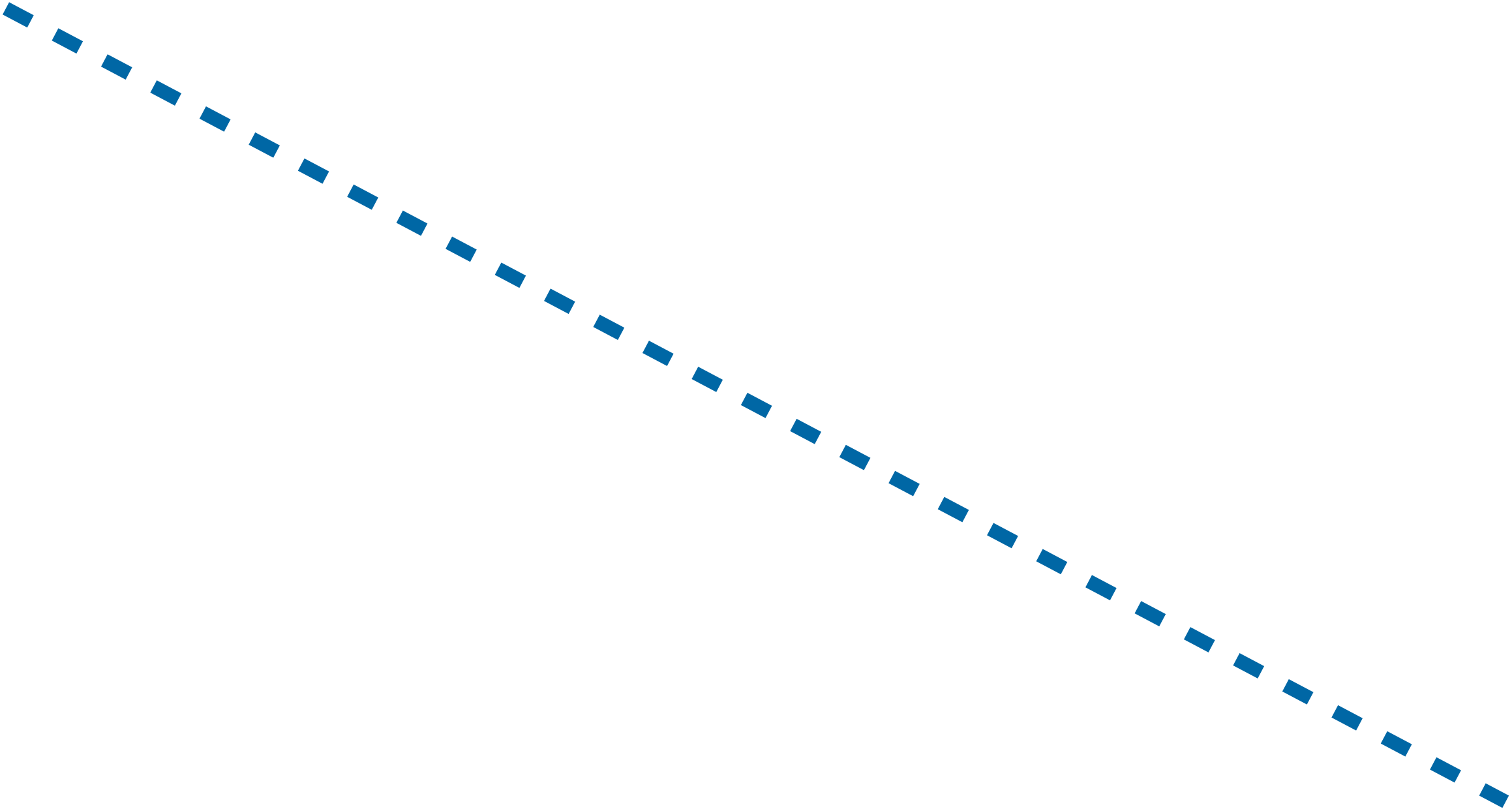


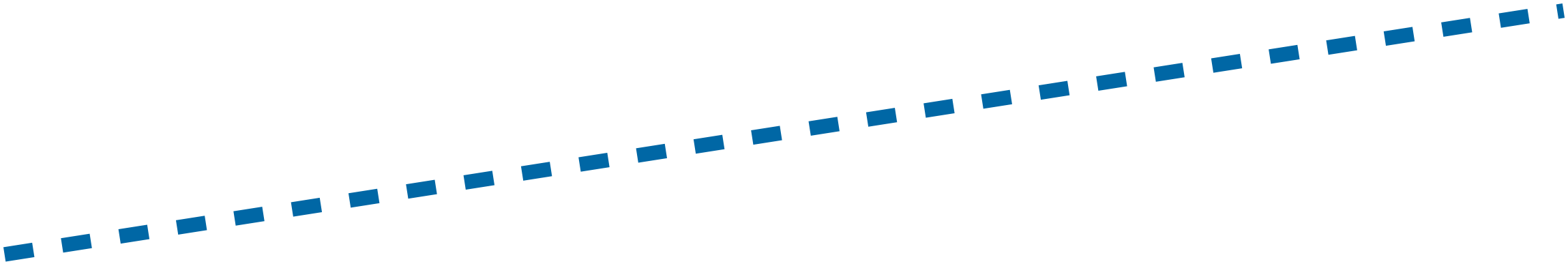
Human Retina

Rhcastilhos & Jmarchn, Schematic diagram of the human eye

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Direction of signals



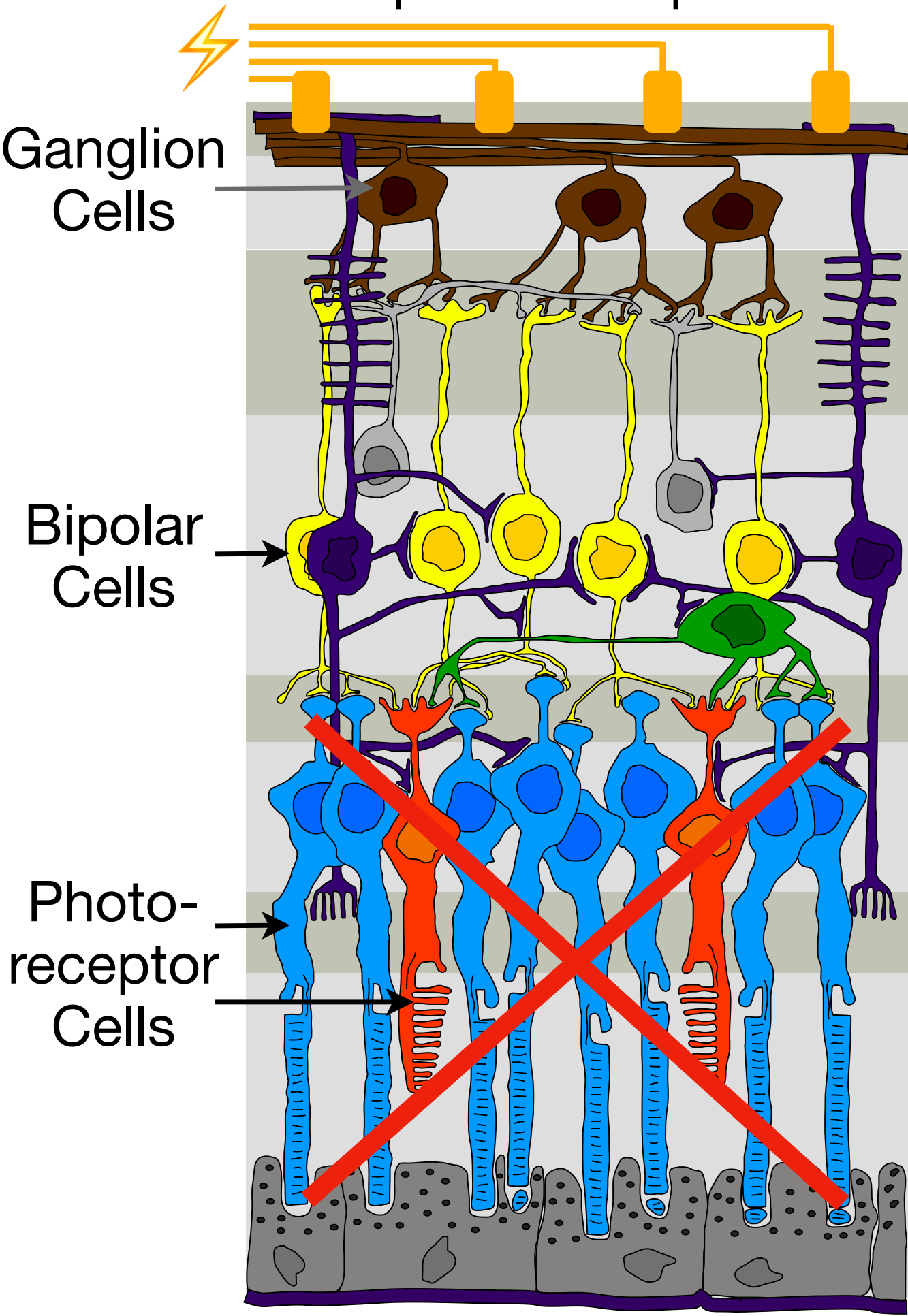
Ganglion
Cells

Bipolar
Cells

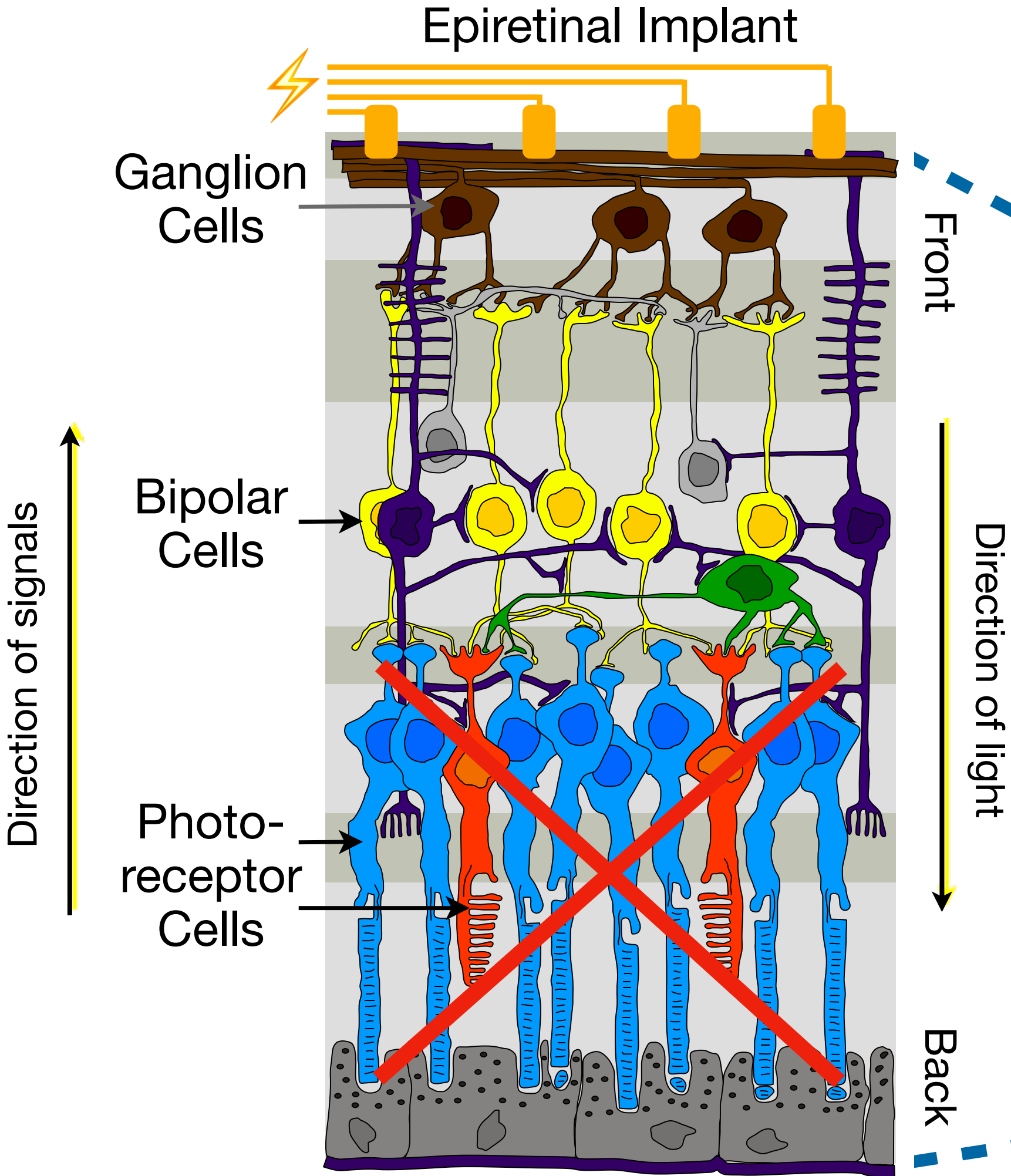
Photo-
receptor
Cells



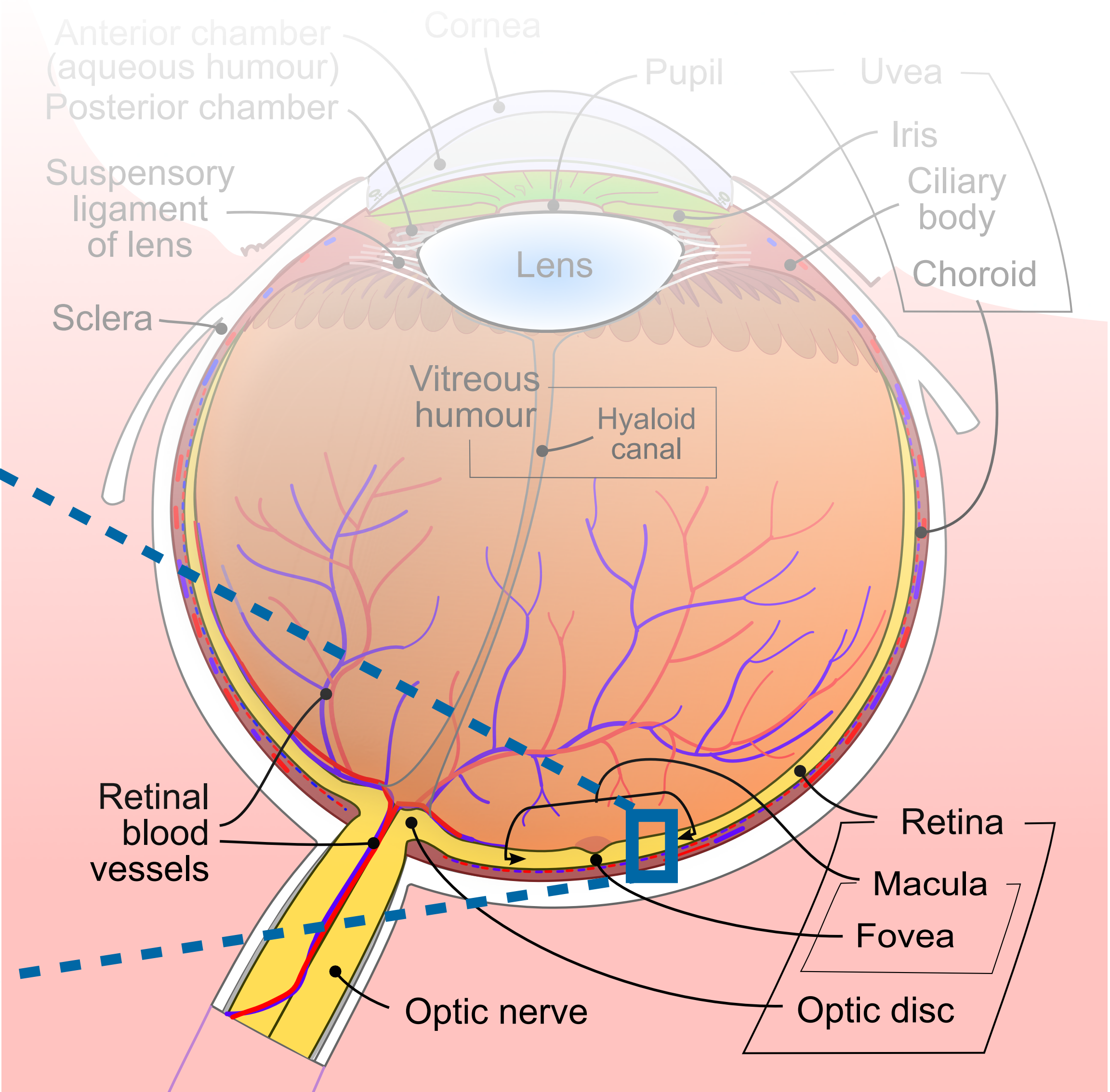
Epiretinal Implant



Human Retina



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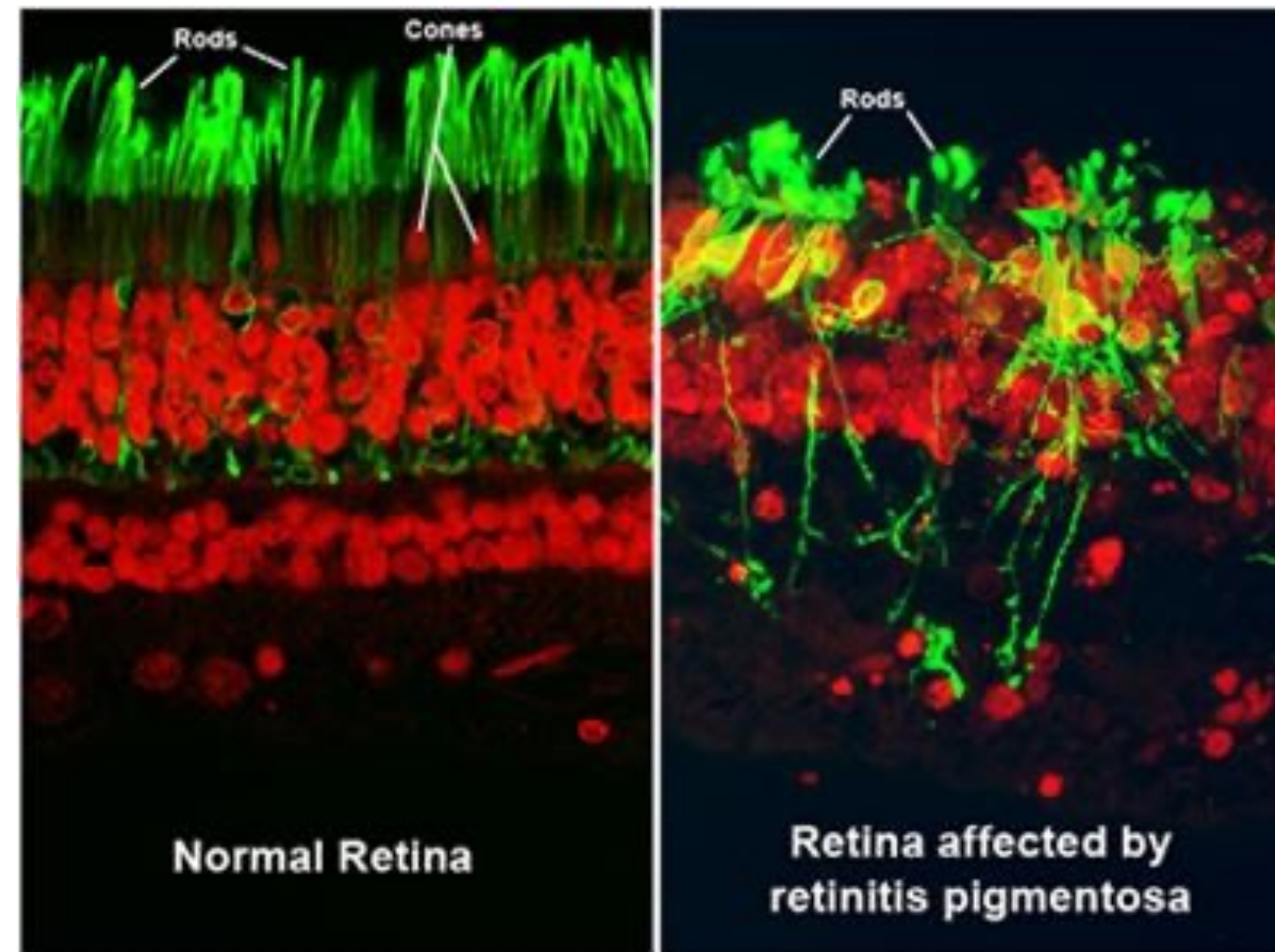


Rhcastilhos & Jmarchn, Schematic diagram of the human eye
CC BY-SA 3.0, via Wikimedia Commons

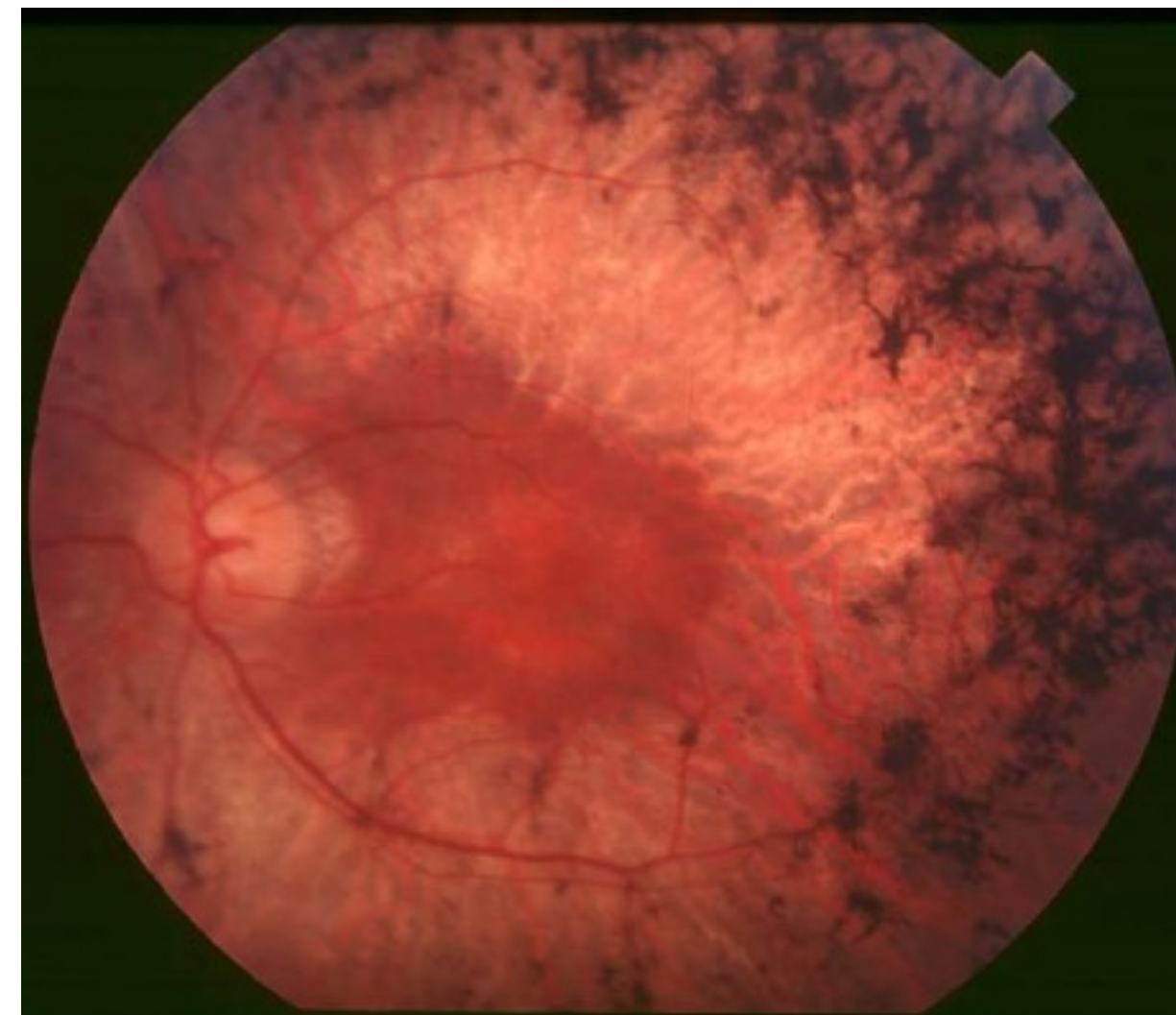
Retinitis Pigmentosa

Hereditary degenerative retinal disease with progressive loss of photoreceptors

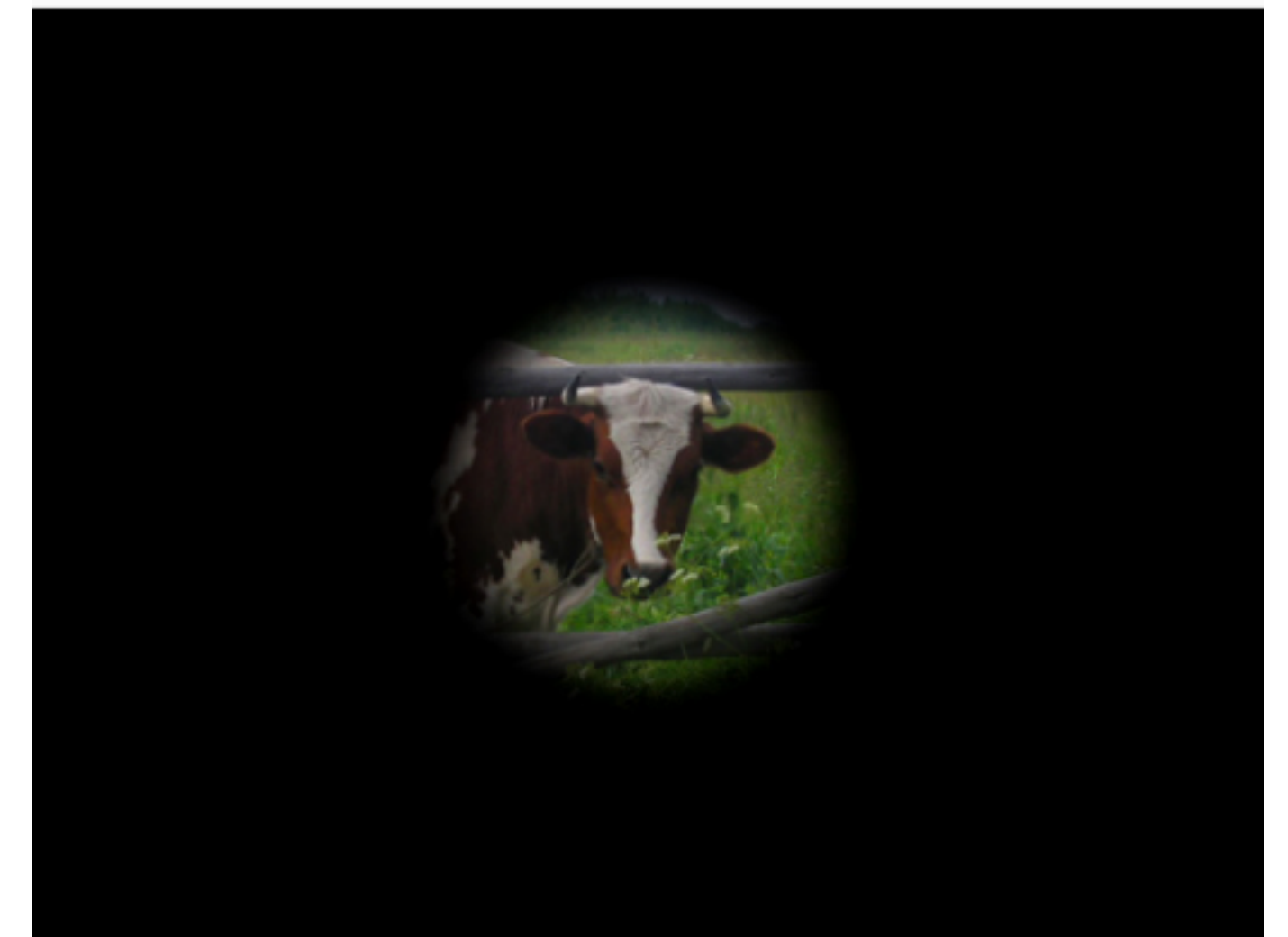
- **Early Symptoms:** Decreased night vision, loss of peripheral (side) vision
- **Late Symptoms:** Vision loss, blindness
- Estimated to affect **1 in 4000** people (1.5 million worldwide) [Hartong et al., 2006]



<https://www.nei.nih.gov/learn-about-eye-health/eye-conditions-and-diseases/retinitis-pigmentosa>



[Hamel, 2006]



Tunnel Vision

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